

Lab 3: Component Modelling & Architectural Pattern Selection

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Architectural Choice

Layered Architecture is selected for the Micro-Lending & Peer Credit Risk Assessor system. The architecture is divided into **Presentation, Business Logic, and Data layers**. The Presentation Layer handles interactions with the Borrower and Peer Lender, the Business Logic Layer handles loan management, credit risk assessment, repayment scheduling, and default warnings, and the Data Layer manages loan-related data. The Lab 3 handout describes Layered Architecture in terms of Presentation, Business, and Data layers.

Reason 1

Layered Architecture provides **clear separation of concerns**. The user interface is separated from business operations such as credit risk assessment and repayment scheduling. This makes the different functions of the system easier to organize and maintain.

Reason 2

The architecture is suitable because the system has distinct functions for **risk assessment, repayment schedules, and automated default warnings**. The credit risk assessment can use the specified factors—debt-to-income ratio, repayment history, and employment stability—to determine Low, Medium, or High risk.

Security Advantage

The layered structure can restrict direct access to financial and loan data by requiring requests to pass through the appropriate business components. In addition, the problem statement

requires loan repayment transactions and EMI distribution to execute using **database ACID transactions to prevent double spending**.

Performance Benefit

Separating loan management, credit risk assessment, repayment scheduling, and default warning functions allows each component to perform a specific task. This can reduce unnecessary processing between system functions and support efficient handling of loan and repayment operations.